

Autodesk® Scaleform®

Memory System Overview

Scaleform 3.0

: Michael Antonov Maxim Shemanarev

: 4.01

: 2011 2 17

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Autodesk® Scaleform® 4.4

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Autodesk Scaleform

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Memory System Overview ()

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

Fax

(301) 446-3199

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1 Scaleform 3.x メモリシステム

Autodesk® Scaleform® 3.0

Scaleform 3.2
AMP Analyzer for Memory and Performance
Scaleform 3.3

1.2

Scaleform 3.3

-
- Movie Scaleform
- ActionScript
-

1.1 メモリのコンセプト

Scaleform

Scaleform

1.1.1 オーバーライド可能なアロケータ

Scaleform

Gfx::System

1. Scaleform::SysAlloc Alloc/Free/Realloc dlmalloc
2. Scaleform Pause Menu

3.
Scaleform

Scaleform::SysAllocWinAPI

(2)

2

1.1.2メモリヒープ

AMP Scaleform
 MemReport

- Global
- MovieData
- MovieView
 ActionScript
- MeshCache

SWF/GFx

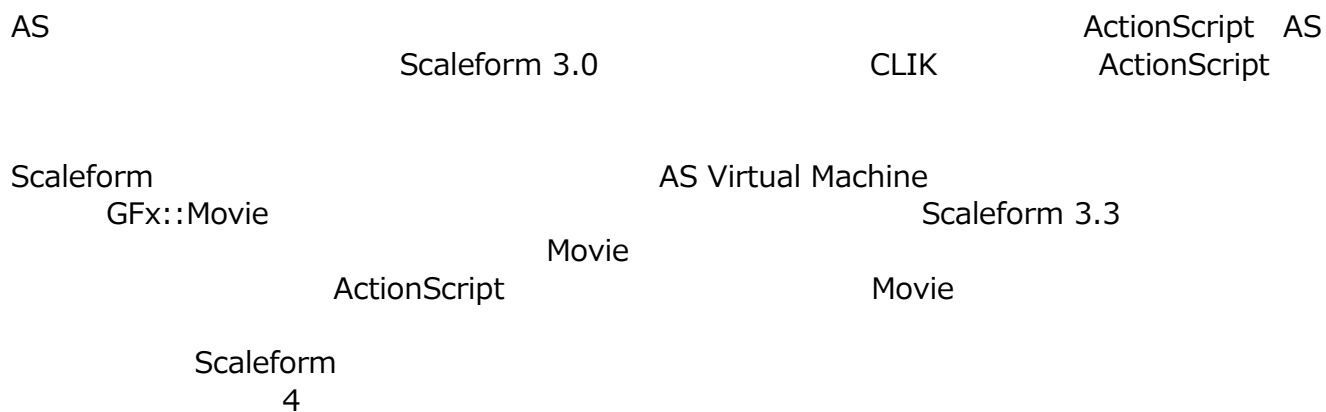
Gfx::Movie

-
-
-
-

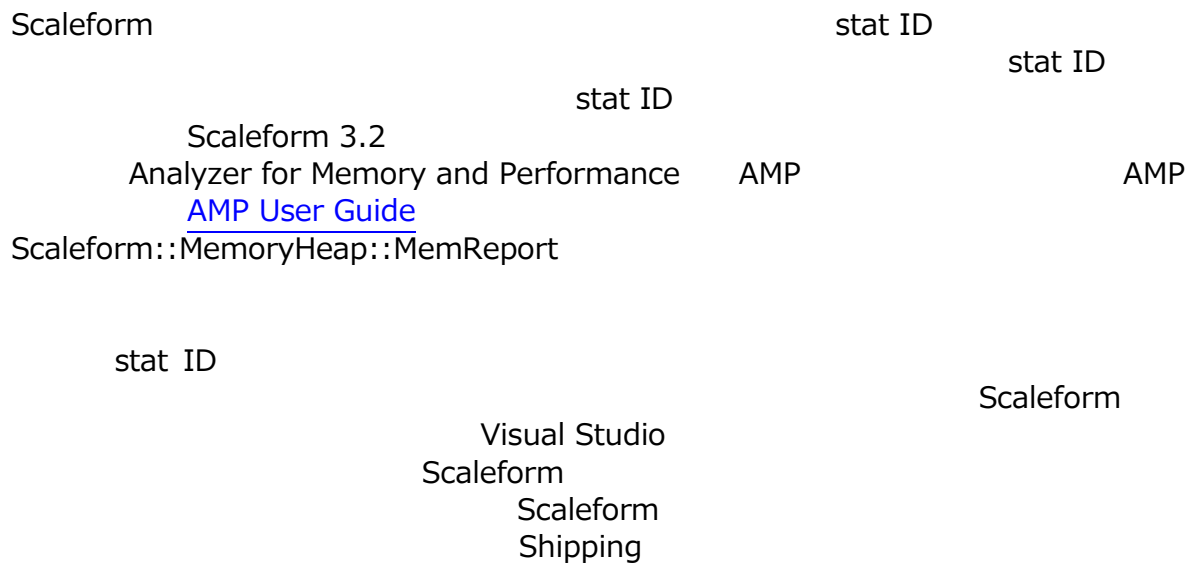
Scaleform 3.3

Scaleform::SysAlloc
 Scaleform 3.3
 Gfx::Movie

1.1.3 ガーベージコレクション



1.1.4 メモリレポートとデバッグ



1.2 Scaleform 3.3 メモリ API の変更

Scaleform 3.3

- Gfx::Movie

- SysAlloc

4K

4

malloc

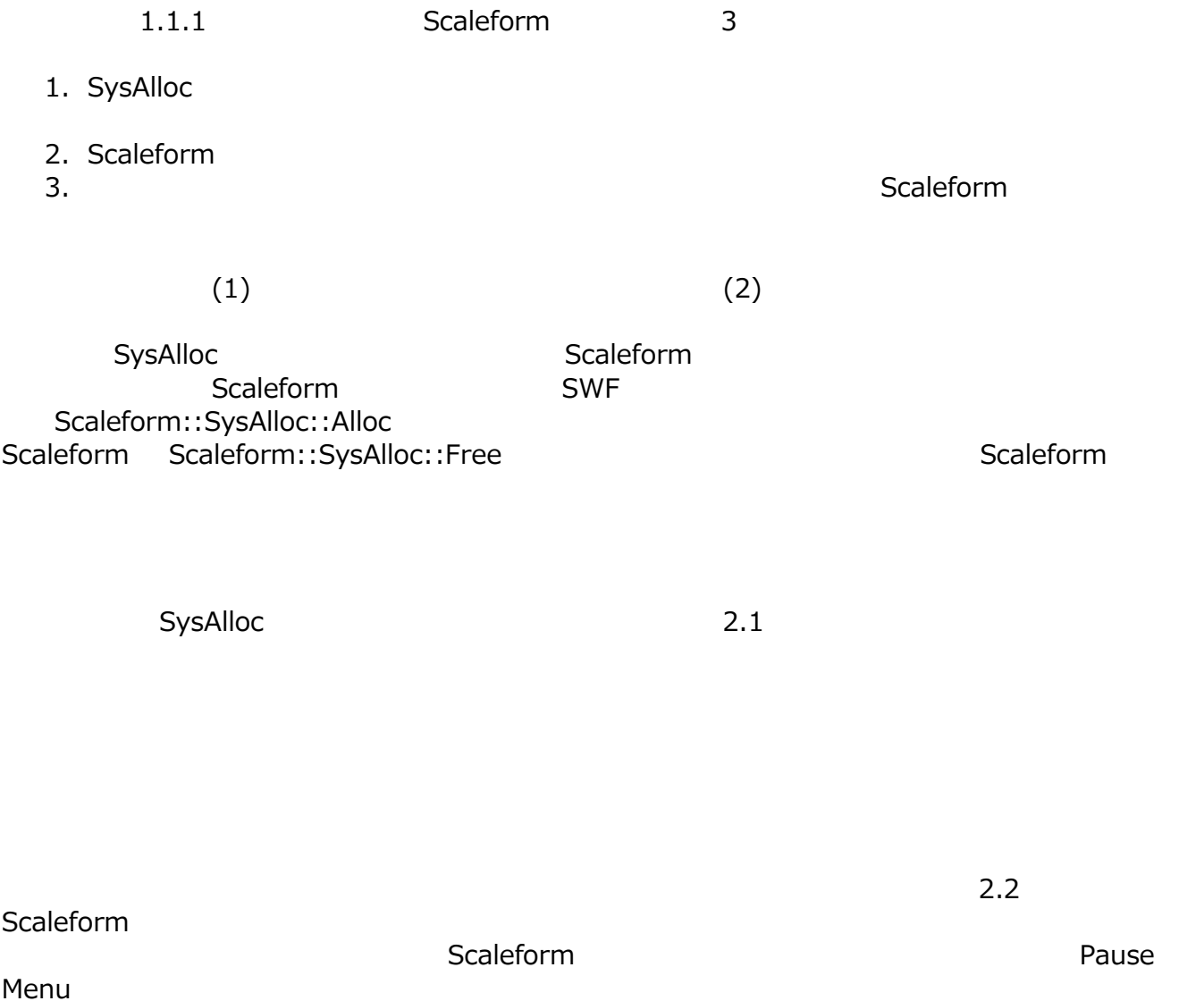
512

Scaleform::SysAlloc
Scaleform::SysAllocPaged

SysAlloc
SysAllocStatic

SysAllocPaged

2 メモリアロケーション



2.1 アロケーションをオーバーライドする



GFx::System

```
Scaleform::SysAllocMalloc
    malloc/free
```

malloc/free

SysAllocMalloc

Scaleform

2.1.1 独自の SysAlloc を実装する

GFx SysAllocMalloc

Windows SysAllocMalloc


```
class MySysAlloc : public SysAlloc
{
public:
    virtual void* Alloc(UPInt size, UPInt align)
    {
        return _aligned_malloc(size, align);
    }

    virtual void Free(void* ptr, UPInt size, UPInt align)
    {
        SF_UNUSED2(size, align);
        _aligned_free(ptr);
        return true;
    }

    virtual void* Realloc(void* oldPtr, UPInt oldSize,
                          UPInt newSize, UPInt align)
    {
        SF_UNUSED(oldSize);
        return _aligned_realloc(oldPtr, newSize, align);
    }
};
```

3 MySysAlloc SysAlloc
 Alloc、Free、Realloc

oldSize

Scaleform	16
align	

Free/Realloc

Alloc/Free

Realloc

Scaleform

Gfx::System

```

MySysAlloc myAlloc;
System gfxSystem(&myAlloc);

```

Scaleform 3.0 Gfx::System Scaleform

Scaleform

Scaleform

(Gfx::System)

 Gfx::System::Init() Gfx::System::Destoy()

 Gfx::System Gfx::System::Init()

SysAlloc

2.2 Scaleform の固定メモリブロックの使用

SysAlloc Scaleform

SysAllocStatic

```

void*      pmemChunk = ...;
GSysAllocStatic      blockAlloc(pmemChunk, 6*1024*1024);
GfxSystem      gfxSystem(&blockAlloc);
...

```

6 Scaleform

 gfxSystem blockAlloc

Scaleform

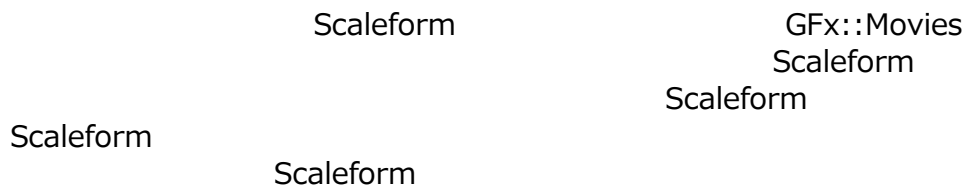
 SysAllocStatic

Alloc 0 Scaleform

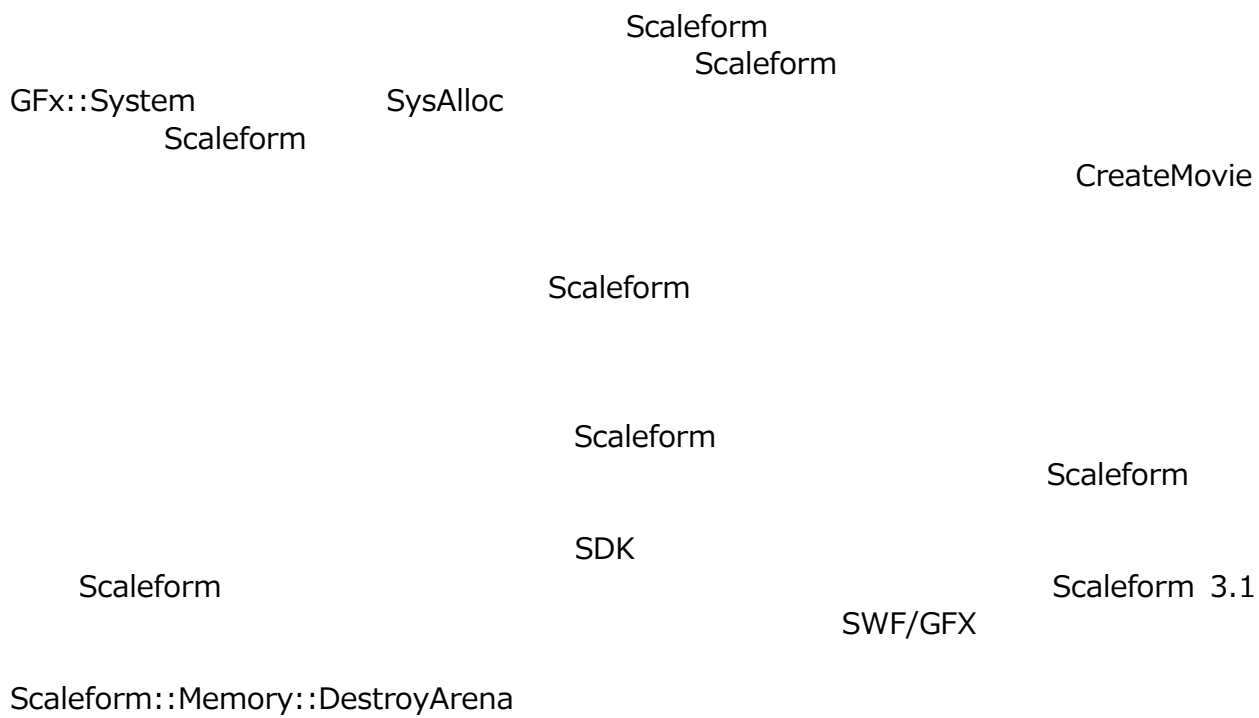
 SysAllocPaged

2.2.1 メモリ アリーナ

Scaleform 3.1



2.2.1.1 背景



2.2.1.2 メモリ アリーナを使用する

•

•

1. Scaleform::Memory::CreateArena

```
(
    ) SysAlloc
    SysAllocStatic
```

```
// pbuf1 10,000,000
SysAllocStatic sysAlloc(pbuf1, 100000000);
Memory::CreateArena(1, &sysAlloc);
```

2. Gfx::Loader::CreateMovie / Gfx::MovieDef::CreateInstance ID

```
// 1
Ptr<MovieDef> pMovieDef =
    *Loader.CreateMovie(filenameStr, Loader::LoadWaitFrame1, 1);
...
// 1
Ptr<Movie> pMovie =
    *pnewMovieDef->CreateInstance(MovieDef::MemoryParams(1), false);
```

3. Gfx::MovieDef/Gfx::Movie

4. Gfx::ThreadedTaskManager Gfx::MovieDef::WaitForLoadFinish Gfx::ThreadedTaskManager

```
pMovieDef->WaitForLoadFinish(true);
pMovie = 0;
pMovieDef = 0;
```

```
(Scaleform::Ptr
Movie->Release();
pMovieDef->Release());
```

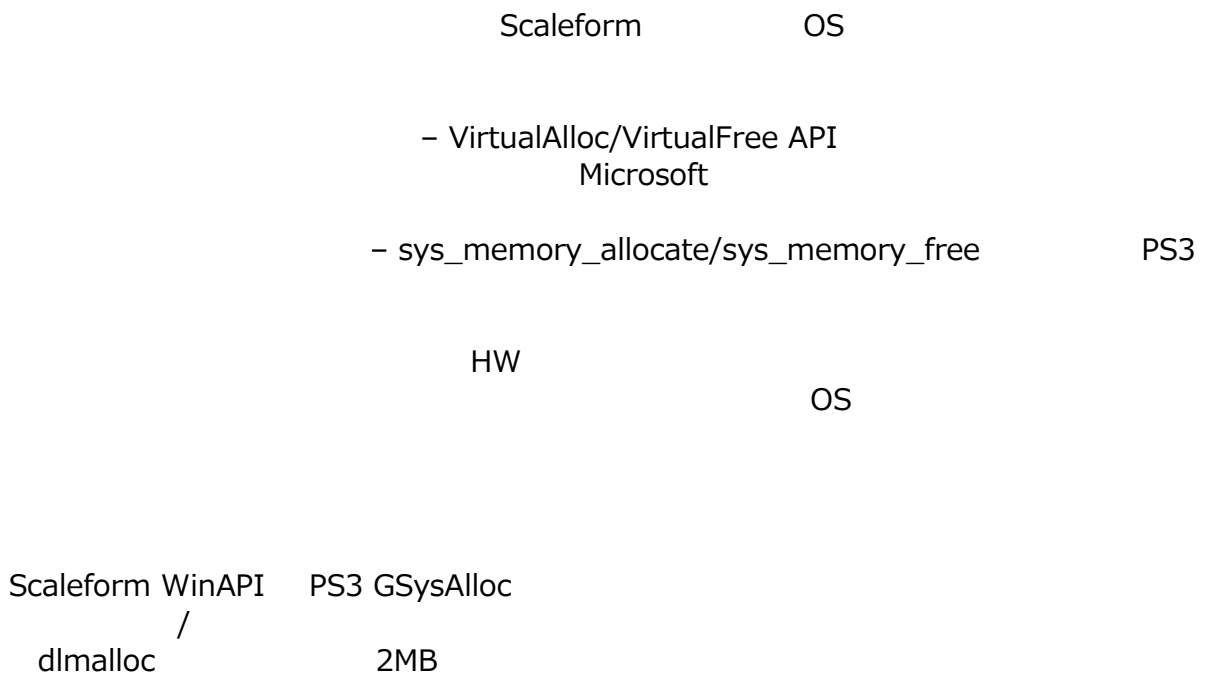
5. DestroyArena CreateArena Scaleform::Memory::ArenaIsEmpty

```
// DestroyArena
```

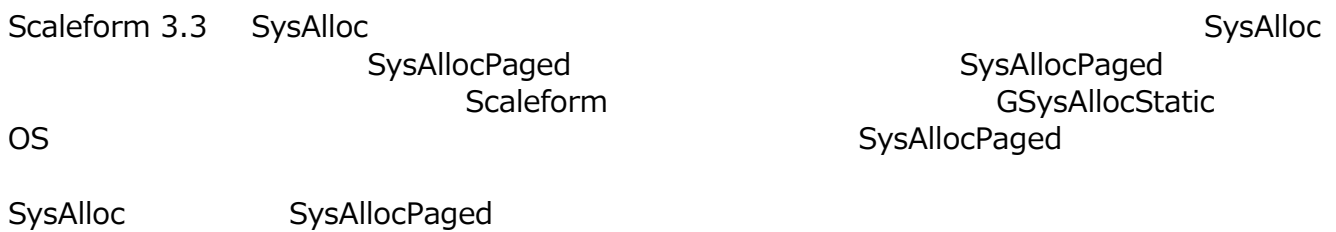
```
// Release    "return *(int*)0;"
Memory::DestroyArena(1);
```

```
SysAlloc
    'pbuff1'
```

2.3 OS へのアロケーションの委任



2.3.1 SysAllocPaged のインプリメンテーション



SysAllocPagedMalloc

:

```
class MySysAlloc : public SysAllocPaged
{
public:
    virtual void GetInfo(Info* i) const
    {
        i->MinAlign    = 1;
        i->MaxAlign    = 1;
        i->Granularity = 128*1024;
        i->HasRealloc  = false;
    }

    virtual void* Alloc(UPInt size, UPInt align)
    {
        // Ignore 'align' since reported MaxAlign is 1.
        return malloc(size);
    }

    virtual bool Free(void* ptr, UPInt size, UPInt align)
    {
        // free() doesn't need size or alignment of the memory block, but
        // you can use it in your implementation if it makes things easier.
        free(ptr);
        return true;
    }
};
```

MySysAlloc
GetInfo、Alloc、Free

GSysAlloc

3
ReallocInPlace

- GetInfo() –
Scaleform::SysAllocPaged::Info
- Alloc –

Alloc

GetInfo MaxAlign
2

MaxAlign 1

- Free – Alloc

Free

Free

- ReallocInPlace –
false

[Scaleform Reference Documentation](#)


```
Gfx::System gfxSystem(&myAlloc);
```

2.4 メモリ ヒープ

Scaleform

API

Scaleform

2.4.1 ヒープ API

'new'

Scaleform

```
Ptr<FileOpener> opener = *new FileOpener();  
loader.SetFileOpener(opener);
```

'new'
Scaleform

Gfx::System

0

と

Gfx::MeshCacheManager

Gfx::Loader::CreateMovie()

SWF/GFX

Gfx::MovieDef::CreateInstance

ActionScript

Scaleform

る

Scaleform

Scaleform

Scaleform::MemoryHeap

Scaleform

Scaleform::MemoryHeap::CreateHeap

Scaleform::MemoryHeap::Release

```
Scaleform::MemoryHeap::Alloc
Scaleform::MemoryHeap::Free
```

SysAlloc

Scaleform

```
Scaleform::MemoryHeap
:
```

- SF_HEAP_ALLOC(heap, size, id)
- SF_HEAP_MEMALIGN(heap, size, alignment, id)
- SF_HEAP_NEW(heap) ClassName
- SF_FREE(ptr)

Scaleform AMP

```
'pHeap'
Gfx::SpriteDef
:
```

```
Ptr<SpriteDef> def = *SF_HEAP_NEW(pHeap) SpriteDef(pdataDef);
```

```
Gfx::SpriteDef Scaleform
Ptr<> Scaleform
```

Release

```
Scaleform::NewOverrideBase
```

```
'delete'
```

```
delete SF_FREE(address)
```

2.4.2自動ヒープ

```
Scaleform::String Scaleform::Array<>
```

```
:
```

```
class Sprite : public RefCountBase<Sprite>
{
    String Name;
    Array<Ptr<Sprite> > Children;
```

GFx::Sprite

Scaleform

- SF_HEAP_ALLOC SF_HEAP_NEW

•

•

'pbuffer'

$$(\quad)$$

Gfx::Sprite

:

```
class Sprite : public RefCountBase<Sprite>
{
    StringLH          Name;
    ArrayLH<Ptr<Sprite> > Children;

    Sprite(String& name, ...) { ... }
};
Ptr<Sprite> sprite = *SF_HEAP_NEW(pHeap) Sprite(name);
sprite->DoSomething();

Scaleform::String          Scaleform::Array<>
Scaleform::StringLH      Scaleform::ArrayLH<>
```

Scaleform

2.4.3ヒープトレーサー

GfxConfig.h SF_MEMORY_TRACE_ALL
 CreateHeap DestroyHeap Alloc Free
Scaleform::Memory SetTracer

SysAllocWinAPI

```
#include "Kernel/HeapPT/HeapPT_SysAllocWinAPI.h"

class MyTracer : public Scaleform::MemoryHeap::HeapTracer
{
public:
    virtual void OnCreateHeap(const MemoryHeap* heap)
    {
        //...
    }

    virtual void OnDestroyHeap(const MemoryHeap* heap)
    {
        //...
    }

    virtual void OnAlloc(const MemoryHeap* heap, UPInt size, UPInt align,
                        unsigned sid, const void* ptr)
    {
```

```

        //...
    }

    virtual void OnRealloc(const MemoryHeap* heap, const void* oldPtr,
                          UInt newSize, const void* newPtr)
    {
        //...
    }

    virtual void OnFree(const MemoryHeap* heap, const void* ptr)
    {
        //...
    }
};

static MyTracer tracer;
static SysAllocWinAPI sysAlloc;
static Gfx::System* gfxSystem;

class MySystem : public Scaleform::System
{
public:
    MySystem() : Scaleform::System(&sysAlloc)
    {
        Scaleform::Memory::GetGlobalHeap()->SetTracer(&tracer);
    }
    ~MySystem() { }
};

SF_PLATFORM_SYSTEM_APP(FxPlayer, MySystem, FxPlayerApp)

```

3 メモリレポート

Scaleform 3.0 Scaleform Player AMP HUD

Scaleform
AMP

[AMP User Guide](#)

Scaleform Player HUD UI

Ctrl F5

Scaleform::MemoryHeap::MemReport

```
void MemReport (class StringBuffer& buffer, MemReportType detailed,  
                bool xmlFormat = false);  
void MemReport (struct MemItem* rootItem, MemReportType detailed);
```

MemReport XML

MemReportType

- MemoryHeap::MemReportBrief – Scaleform

Memory	4,680K / 4,737K
Image	1,052
Sound	136
Movie View	1,739,784
Movie Data	300,156

- MemoryHeap::MemReportFull – Ctrl F6 Scaleform Player

SF_MEMORY_ENABLE_DEBUG_INFO stat ID

Memory	4,651K / 4,707K
System Summary	
System Memory FootPrint	4,832,440
System Memory Used Space	4,775,684
Debug Heaps Footprint	12,884
Debug Heaps Used Space	5,392
Summary	
Image	1,052
Sound	136
Movie View	1,715,128

Movie Data	300,156
[Heap] Global	4,848,864
Heap Summary	
Total Footprint	4,848,864
Local Footprint	2,427,860
Child Footprint	2,421,004
Child Heaps	4
Local Used Space	2,417,196
DebugInfo	120,832
Memory	
MovieDef	
Sounds	8
ActionOps	80,476
MD_Other	200
MovieDef	36
MovieView	
MV_Other	32
General	91,486
Image	92
Sound	136
String	31,425
Debug Memory	
StatBag	16,384
Renderer	
Buffers	2,097,152
RenderBatch	5,696
Primitive	1,512
Fill	900
Mesh	4,884
MeshBatch	2,200
Context	15,184
NodeData	15,160
TreeCache	13,220
TextureManager	2,336
MatrixPool	4,096
MatrixPoolHandles	2,032
Text	1,232
Renderer	13,488
Allocations Count	1,822

- MemoryHeap:: MemReportFileSummary –
SWF MovieData MovieDef

MovieView statID

Movie File 3DGenerator_AS3.swf
Memory 1,956,832

MovieDef	152,664
CharDefs	39,848
ShapeData	1,716
Tags	73,656
MD_Other	37,444
MovieView	196,249
MovieClip	120
ActionScript	181,917
MV_Other	14,212
General	1,596,712
Image	960
String	2,783
Renderer	7,464
Text	7,464

- MemoryHeap::MemReportHeapDetailed – AMP Memory Scaleform

Total Footprint	4,858,148
Used Space	4,793,972
Global Heap	2,405,492
MovieDef	80,720
Sounds	8
ActionOps	80,476
MD_Other	200
MovieView	32
MV_Other	32
General	92,566
Image	92
Sound	136
String	31,276
Debug Memory	8,192
StatBag	8,192
Renderer	2,182,960
Buffers	2,097,152
RenderBatch	5,696
Primitive	1,512
Fill	900
Mesh	4,884
MeshBatch	2,200
Context	15,184
NodeData	15,016
TreeCache	13,136
TextureManager	2,336
MatrixPool	8,192
MatrixPoolHandles	2,032

Text	1,232
Movie Data Heaps	300,156
MovieData "3DGenerator_AS3.swf"	300,156
MovieDef	152,664
CharDefs	39,848
ShapeData	1,716
Tags	73,656
MD_Other	37,444
General	141,332
Image	960
String	2,312
Movie View Heaps	1,727,936
MovieView "3DGenerator_AS3.swf"	1,727,936
MovieView	195,977
MovieClip	120
ActionScript	181,645
MV_Other	14,212
General	1,467,668
String	471
Renderer	7,464
Text	7,464
Other Heaps	360,580
_FMOD_Heap	360,580
General	359,944
Debug Data	12,884
Unused Space	51,292
Global	51,292
_FMOD_Heap	4,384
MovieData "3DGenerator_AS3.swf"	5,356
MovieView "3DGenerator_AS3.swf"	33,032

Scaleform	SF_MEMORY_ENABLE_DEBUG_INFO
	stat ID

- MovieDef –
- MovieView – Gfx::Movie – ActionScript – ActionScript
- CharDefs –
- ShapeData –
- Tags –
- ActionOps – ActionScript
- MeshCache – EdgeAA

- FontCache -

4 ガーベージ コレクション

Scaleform 2.2 ActionScript

ActionScript 2

1

Code:

```
var o1 = new Object;
var o2 = new Object;
o1.a = o2;
o2.a = o1;
```

K

%

```
#define GFX_AS_ENABLE_GC
```

Scaleform

4.1 ガーベージ コレクションを使えるようにメモリ ヒープを構成する

Scaleform 3.x

Scaleform 3.3

MovieView

MovieDef

```
Movie* MovieDef::CreateInstance(const MemoryParams& memParams,  
                                bool initFirstFrame = true)
```

MemoryParam

```
struct MemoryParams  
{  
    MemoryHeap::HeapDesc Desc;  
    float HeapLimitMultiplier;  
    unsigned MaxCollectionRoots;  
    unsigned FramesBetweenCollections;  
};
```

- Desc –
MinAlign) 、 (Granularity) Limit) Flag)
Scaleform
- HeapLimitMultiplier – 0…1
- MaxCollectionRoots –
- FramesBetweenCollections –

movie view

```
MemoryContext* MovieDef::CreateMemoryContext(
```

```
const char* heapName,
const MemoryParams& memParams,
bool debugHeap )
```

```
Movie* MovieDef::CreateInstance(MemoryContext* memContext,
                                bool initFirstFrame = true)
```

```
MemoryContext          movie view          movie view
                        AMP                  AMP
```

movie view

```
MemoryParams::Desc      ActionScript
HeapLimitMultiplier     Desc.Limit と
```

```
128K          "          "
```

Boehm-Demers-Weiser(BDW)

BDW

```
if (allocs since collect >= heap footprint * HeapLimitMultiplier)
    collect
else
    expand(heap footprint + overlimit + heap footprint *
          HeapLimitMultiplier)
```

```
"collect"      ActionScript
```

```
HeapLimitMultiplier    0.25
                        25
HeapLimitMultiplier    Scaleform
Scaleform              25
```

```
Desc.Limit
Desc.Limit
```

```

MaxCollectionRoots/
FramesBetweenCollections
MaxCollectionRoots
ActionScript
ActionScript
ActionScript
ActionScript
MaxCollectionRoots
MaxCollectionRoots
"obj"
"obj.member=1"
0
FrameBetweenCollections
" "
30fps
Advance
Flash 1 30
SWF 2 60
1800
60
Desc.Limit
Flash
FrameBetweenCollections
ActionScript
0

```

4.2 Gfx::Movie:: ForceCollectGarbage

```
virtual void ForceCollectGarbage() = 0;
```